A squared-loglog upper bound for the comparability partition number of the interval graph G_n

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Editorial note. The counted unit is the article's own Theorem ub at source lines 468-470 together with the author's own proof environment at source lines 609-618, whose header is Proof of Theorem ub; the proof is not adjacent to the statement, because the author places the preparatory definition of F_n at source lines 475-479, the recurrence Lemma rec with its long proof at source lines 484-598 and the corollary p' at source lines 602-605 in between, and the pairing therefore rests on the author's own named header rather than on a later-first-proof rule.

Editorial note. The retained context at source lines 128-129, 391-395, 409-411, 475-479 and 602-605 is the article's own definitions of the parameters p(G) and c(G), of the interval graph G_n and of the numbers c(n), p(n), F_n and p'(n), together with two of its own corollaries: the bound for the partition number of the shift graph S_n and the quadratic bound for p'(n) at powers of two. Both corollaries are retained as quoted prerequisites of the counted proof, not as counted units.

Editorial note. The retained context at source lines 416-419 and 602-605 carries no citation and no cross-reference. The body of Lemma rec at source lines 484-598, which is the interval that is publicly occupied for this source, is deliberately not retained.

Editorial note. The source preamble cannot be retained as it stands: it loads rawfonts, psfrag, verbatim, tikz with several libraries, hyperref with the pagebackref option and a dvi driver option for graphicx, none of which

the retained body spans use, and it redefines backref commands that only exist because of that hyperref option. The retained preamble file therefore carries, line for line, source line 2, source line 3, source line 4, source line 16, source line 17, the theorem declarations of source lines 20-30 and the author macro definitions of source lines 32-41; the only textual change anywhere is the removal of the dvipdfm driver option from the graphicx line, because the assembly forces the xelatex engine. No new macro, package or declaration was introduced.

Editorial note. The article's theorems co-triangulated, unit, gensplit, bipline and loglog, its lemma rec, its corollaries doblesh and shift as counted statements, and its proposition H9 are not counted here.

Allowed standard prerequisites (editorial).

- The partition number p(G) and the cover number c(G). p(G) is the minimum m such that the edge set of G is the union of m edge-disjoint comparability subgraphs of G, and c(G) is the same number without the edge-disjointness requirement; retained at source lines 128-129.. The counted theorem is an upper bound for p(n)=p(G_n), so the parameter and its edge-disjointness requirement must be fixed before the statement can be read.

- The interval graph G_n and the number p(n). For n at least 2, G_n is the intersection graph of the n choose 2 closed intervals whose endpoints are among n consecutive positive integers, and the article abbreviates c(n)=c(G_n) and p(n)=p(G_n); retained at source lines 391-395.. Identifies the graph whose partition number the counted theorem bounds, and shows that p(n) is non-decreasing in n, which the proof uses to reduce to the powers of two.

- The shift graph S_n and the bound on its partition number. S_n is the subgraph of the touching edges of G_n, its chromatic number equals the ceiling of the logarithm of n, and the article records at source lines 416-419 the corollary that p(S_n) is at most the ceiling of the logarithm of the ceiling of the logarithm of n, with a partition into bipartite graphs; retained at source lines 409-411 and 416-419.. The proof splits the edge set of G_n into the touching part S_n and the rest F_n and quotes exactly this bound for the first part.

- The graph F_n and the number p'(n). F_n is obtained from G_n by deleting all touching edges without deleting their endpoints, and p'(n) is defined as p(F_n); retained at source lines 475-479.. The proof uses that the edge set of G_n is the disjoint union of the edge sets of S_n and F_n, which is what allows the two partitions to be combined.

- The corollary bounding p'(n) at the powers of two. For every n of the form 2^{2^k} with k at least 1 the article proves the chain of inequalities p'(n)=p'(2^{2^{k-1}}) <= p(2^{2^{k-1}})+k-1+2 <= k choose 2 + 2k - 1; retained at source lines 602-605.. The proof substitutes exactly this quadratic bound for p'(n) at n=2^{2^k} to obtain the stated estimate.

Let $p(G)$ denote the minimum number m such that $E(G) = \cup_{i=1}^{m} E(G_i)$ where the G_i-s are edge-disjoint comparability subgraphs of G. Dropping the condition that the G_i-s are edge-disjoint, we define $c(G)$ as the smallest number of comparability subgraphs of G needed to *cover* $E(G)$.

For $n \geq 2$, let G_n be the interval graph defined as the intersection graph of the $\binom{n}{2}$ closed intervals with endpoints among n consecutive positive integers, called *base points*. Note that $|V(G_n)| = \binom{n}{2}$. For convenience, set $c(n) = c(G_n), p(n) = p(G_n)$. Since G_2 has one vertex and no edge, we set $c(2) = p(2) = 0$.

Let S_n denote the subgraph of touching edges of G_n. The graph S_n is often referred as the *shift graph*. It is well-known that $\chi(S_n) = \lceil \log(n) \rceil$

Corollary 1. *$p(S_n) \leq \lceil \log(\chi(S_n)) \rceil = \lceil \log(\lceil \log(n) \rceil) \rceil$ and the partition can be done by bipartite graphs.*

Before the proof, we need some preparations. Let F_n be the graph obtained from G_n by removing all touching edges of G_n, i.e. edges of S_n (without removing their endpoints). Let $p'(n) = p(F_n)$.

Corollary 2. *For any $n = 2^{2^k}$ with $k \geq 1$ we have*

$$p'(n) = p'(2^{2^{k-1}}) \leq p(2^{2^{k-1}}) + k - 1 + 2 \leq \cdots \leq \binom{k}{2} + 2k - 1.$$

Theorem 3. *For any $n \geq 4$ we have*

$$p(n) \leq \frac{1}{2}(\lceil \log(\lceil \log(n) \rceil) \rceil)^2 + \frac{5}{2} \lceil \log(\lceil \log(n) \rceil) \rceil - 1.$$

Proof of Theorem 3. It is enough to prove the statement for $n = 2^{2^k}$ where $k \geq 2$. Since $p(n)$ is non-decreasing, this implies the statement for $2^{2^{k-1}} < n < 2^{2^k}$ as well.

Let $n = 2^{2^k}$. Observe that the edge set of G_n is the disiont union of the edge set of S_n and F_n. Combining the partitions given by Corollary 1 and 2 respectively we get that

$$p(n) \leq p'(n) + p(S_n) \leq \binom{k}{2} + 2k - 1 + \lceil \log(\lceil \log(n) \rceil) \rceil = \binom{k}{2} + 3k - 1$$

$$\leq \frac{1}{2}(\lceil \log(\lceil \log(n) \rceil) \rceil)^2 + \frac{5}{2}\lceil \log(\lceil \log(n) \rceil) \rceil - 1.$$

finishing the proof of Theorem 3. □